

AI-Driven Virtual Assistant for INGRES

Problem Statement: SIH25066 | Ministry of Jal Shakti / CGWB | Complete Guide

1. Executive Summary

This document serves as the foundational reference and master guide for our project in the Smart India Hackathon 2026. The challenge focuses on developing an intelligent, AI-powered conversational virtual assistant for the **IN-GRES** (INdia-Groundwater Resource Estimation System) platform created by the Central Ground Water Board (CGWB) and IIT Hyderabad. Our solution transforms over 500 pages of dense government hydrology reports, 36 state fact sheets, and 6,635 granular block-level records into an accessible, interactive, and multilingual decision-support assistant for farmers, industries, citizens, and policymakers.

2. The Real-World Groundwater Crisis in India

India is the largest consumer of groundwater globally, extracting more than 240 Billion Cubic Meters (BCM) annually—surpassing the combined extraction of China and the United States. Groundwater accounts for over **85% of India's rural and urban domestic drinking water** and more than **60% of total agricultural irrigation**. However, unsustainable extraction practices, erratic monsoon patterns, and rapid urbanization have pushed hundreds of blocks into severe groundwater depletion.

3. Plain-English Glossary of Hydrogeological Terms

Understanding the technical terminology used by the Ministry of Jal Shakti is vital. Below is a simplified translation using a banking analogy:

Technical Term	Formal Definition (CGWB)	Bank Account Analogy
Annual Recharge	Total volume of water naturally refilling the underground aquifers each year from rainfall, canals, and surface water bodies.	Annual Salary / Deposits into your bank account.
Extractable Resource	The portion of total recharge safely available for human extraction after subtracting natural environmental discharges (streams, base flow).	Available Balance after mandatory taxes/savings.
Annual Extraction	Total water pumped out of the ground via tube-wells and borewells for irrigation, drinking, and industrial use.	Total Spending / Withdrawals from your account.
Stage of Extraction (SoE %)	$(\text{Total Annual Extraction} / \text{Annual Extractable Resource}) \times 100$. Key metric of aquifer stress.	Spending Ratio : Spending more than your salary means bankruptcy.
BCM	Billion Cubic Meters (1 BCM = 1,000,000,000,000 liters). Standard unit of national water estimation.	The currency denomination (e.g., Crores).
Assessment Unit (Block)	Administrative sub-district unit (Block, Taluk, Mandal, Firka) evaluated as a distinct hydrological unit.	Individual branch/account location.

4. The 4 Official Categorization Tiers (GWRA-2025)

Under the Ground Water Resource Estimation Committee (GEC-2015) guidelines, every assessment unit in India is categorized into four distinct zones based on its Stage of Groundwater Extraction (SoE %):

Category	Extraction Threshold	Current National Count	Regulatory & Physical Implications
SAFE	$\text{SoE} \leq 70\%$	4,945 Blocks (74.5%)	Aquifer is replenishing faster than extraction. New borewells permitted easily.
SEMI-CRITICAL	$70\% < \text{SoE} \leq 90\%$	759 Blocks (11.4%)	Approaching sustainable limit. Water conservation & monitoring mandatory.
CRITICAL	$90\% < \text{SoE} \leq 100\%$	201 Blocks (3.0%)	Severe stress. Extraction equals total replenishable recharge. High alert.
OVER-EXPLOITED	$\text{SoE} > 100\%$	730 Blocks (11.0%)	Severe Crisis Zone. Water tables plummeting rapidly. Strict CGWA NOC clearance required.

5. What is the INGRES System?

INGRES (India-Groundwater Resource Estimation System) is a centralized, GIS-based web application developed jointly by the Central Ground Water Board (CGWB) and the Indian Institute of Technology Hyderabad (IIT-H). It automates the nationwide assessment of dynamic groundwater resources using the GEC-2015 methodology. It incorporates hydrological boundaries, rainfall data, telemetry sensor logs from Digital Water Level Recorders (DWLRs), and spatial GIS layers across states.

6. The Problem Statement: Why INGRES Needs an AI Assistant

While INGRES holds comprehensive scientific water data, its current interface is heavily technical, static, and inaccessible to the people who need it most:

- **High Cognitive Barrier:** Data is presented through dense GIS layers and voluminous PDF reports that require hydrogeological expertise to interpret.
- **No Conversational Access:** Citizens and farmers cannot type a simple question like *'Can I dig a borewell in my village?'* or *'Is water level rising in my district?'*.
- **Lack of Decision-Support for Officials:** Administrators (District Collectors, Panchayats) lack an instant synthesis tool to identify high-stress blocks needing urgent rainwater harvesting funds.
- **Zero Multi-Lingual & Voice Support:** Non-English speakers in rural farming belts cannot interact with the national water platform.

7. Our Proposed AI Virtual Assistant Solution

We are building a full-stack, AI-driven conversational intelligence system that acts as the virtual front-door to INGRES. It combines natural language processing, structured SQL retrieval against 6,635 blocks, RAG-powered knowledge extraction from 36 state fact sheets, and real-time data visualization.

Layer	Component	Function & Capabilities
User Interface	Modern Web & Mobile App (React / Next.js / Vite)	Conversational chat UI with dark mode, interactive Leaflet GIS maps, chart cards, and voice input in Hindi & English.
AI / NLU Engine	Query Router & Function Calling (Gemini LLM)	Translates plain-English/Hindi user intents into precise database queries and synthesizes intuitive, empathetic advisories.
Database Layer	Unified Master DB (SQLite / PostgreSQL)	High-speed indexed storage of 36 state summaries, 6,635 block categorizations, 141 water quality records, and 61 depth trends.
Knowledge RAG	State Fact Sheet Semantic Corpus	Full-text vector search across all 36 state reports for local contamination advisories (Fluoride, Arsenic, Salinity) and depth trends.
INGRES API Spec	Standardized 7-Endpoint Schema	Implements official INGRES REST endpoints: get_country_summary, get_state_data, get_district_data, search_location.

8. Real-World User Scenarios (Persona Walkthrough)

Scenario A: The Farmer (Local Guidance)

'I am a paddy farmer in Sangrur, Punjab. Is groundwater sufficient for next season?'

AI Assistant Response: Alerts the farmer that Sangrur's Stage of Extraction is at 156.8% (Over-Exploited). Displays a visual donut chart showing agricultural water stress and advises on government subsidies for drip irrigation and direct seeded rice.

Scenario B: The Factory Owner (Regulatory / NOC Advisory)

'Can I get a CGWA NOC to extract 50,000 liters/day for a food plant in Bilara, Jodhpur?'

AI Assistant Response: Identifies Bilara block as Over-Exploited. Clearly explains CGWA mandatory rainwater recharge obligations, artificial recharge requirements, and required NOC documentation.

Scenario C: The District Collector (Policy & Resource Allocation)

'Show me all critical and over-exploited blocks in my state for emergency Jal Shakti funding.'

AI Assistant Response: Instantly queries the 6,635 block database, generates an interactive color-coded GIS map, and provides a downloadable CSV/PDF priority report.

9. Current Master Data Lake Status in Project

All project data has been downloaded, parsed, and consolidated into **data/processed/ingres_master.db** and **data/processed/state_factsheets_corpus.json**:

- **36 State & UT Complete Summaries** (Monsoon/Non-monsoon recharge, natural discharge, sector extraction, SoE %).
- **6,635 Individual Block Categorizations** parsed from the official 97-page CGWB 2025 document.
- **141 State Water Quality Contamination Records** (Fluoride, Arsenic, Nitrate, Uranium, EC, Iron).
- **61 Seasonal Water Table Depth Trends** (Pre-Monsoon vs Post-Monsoon ranges and decadal shifts).
- **36 Full State Fact Sheet Texts** indexed for semantic RAG search.
- **8 Major Metropolitan Cities** groundwater extraction and recharge datasets.

10. Hackathon Implementation Plan

1. **Backend:** Build FastAPI / Express server exposing the 7 INGRES endpoints directly connected to our unified SQLite master DB.
2. **AI Layer:** Connect LLM function-calling to route queries to SQL (numerical lookups) or RAG (fact sheet advisories).
3. **Frontend:** Build a responsive dashboard featuring conversational chat, dynamic charts (Recharge vs Extraction), and interactive Leaflet map overlays.
4. **Verification & Polish:** Test multi-turn conversations, voice input, and multi-lingual prompt handling.